Lesson plan · for the teacher

How a network learns

AI · Level 3 · Lesson 2

Time	Materials
40 min	Board and chalk, and one piece of chalk to hide; Space to move between the desks (or the playground)

Learning goals

- Explain how layers of neurons combine straight lines into shapes one neuron can't learn.
- Describe training as a loop: guess, measure the error, nudge every weight a little downhill, repeat.
- Explain why the step size and enough varied examples matter.

Activity: Warmer, colder

1. Warm-up (5 min): Recap lesson 1 with the umbrella neuron: inputs, weights, bias, total. Ask: who sets the weights in a network with billions of them?
2. Layers (8 min): Draw a temperature line from cold to hot. Show that one cut gives "too cold" or "not too cold", and that two cuts, from two neurons, fence in "just right". Explain that hidden neurons draw the lines and the next neuron combines them.
3. Warmer, colder (12 min): One pupil steps out; the class hides a piece of chalk. Back inside, the pupil moves only when the class says "warmer" or "colder". Count the steps it takes. Play twice with different pupils.
4. Big steps, small steps (8 min): Replay with giant steps only (the seeker overshoots and swings back and forth), then with tiny shuffles (slow). Ask which step size worked best, and why.
5. Wrap-up (7 min): Write the training loop on the board: guess, measure the error, nudge, repeat. Match it to the game: the distance to the chalk was the error; "warmer" or "colder" told the seeker which way shrinks it, as the slope tells a network; a step was a nudge; and the step size is the learning rate. Then give the exit ticket.

Choose a hiding place that needs about ten normal steps to reach. Agree the rule: the class says one word after every step, and only "warmer" or "colder".

Discussion

- In the game, what was the error, what told you which way to go, and what was a step?
- Why did giant steps make it harder to find the chalk?
- Which examples would a network need to work well in our town?

Exit ticket

- Write the four steps of the training loop, and one reason the step size matters.

Try it at school tomorrow

Play "Warmer, colder": one pupil steps outside while the class hides a piece of chalk. Back inside, the pupil moves only by the class's "warmer" (closer) or "colder" (further). Play again with giant steps, then with tiny ones. Then ask: what was the error, what was a step, and why did giant steps and tiny steps both cause trouble?

Practice questions · for you

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1 **Which of these needs at least one hidden layer?**

- ◯ Two groups of dots, far apart
- ◯ Circles inside a ring of squares
- ◯ Dots above a slanted line, and dots below it

2 **Put the training loop for one example in order.**

- ☐ Measure the error against the label
- ☐ Show the network a labelled example
- ☐ The network gives its guess
- ☐ Move on to the next example
- ☐ Nudge every weight a little to shrink the error

3 **A leaf-disease app does well on the builders' photos but badly on farms near your school. What should its makers do? Choose all that apply.**

- ☐ Add photos from local farms, taken on local phones
- ☐ Use a bigger learning rate
- ☐ Check its score on local photos it never trained on
- ☐ Add photos in different light and weather
- ☐ Train it for longer on the same photos

4 **Complete the sentence.**

Each round, the network compares its guess with the _____ to get the _____,

then nudges every _____ the way that makes the error _____.

Word bank: weight · rule · error · label · smaller · bigger

✎ **When did you last learn something from "warmer, colder" feedback rather than a rule? How could you give pupils that kind of feedback?**

Check your answers

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1 **Circles inside a ring of squares**

No straight line can separate an inside from a ring around it, so the network needs a hidden layer to draw several lines for the next neuron to combine.

2 **Top to bottom: 3, 1, 2, 5, 4**

The network can only be corrected after it has guessed and the error is measured.

3 **Add photos from local farms, taken on local phones; Add photos in different light and weather; Check its score on local photos it never trained on**

The app learned the hill its photos made: local, varied photos change that hill, and a test on new local photos shows whether it worked.

4 **Each round, the network compares its guess with the label to get the error, then nudges every weight the way that makes the error smaller.**

The label is the right answer; the gap between it and the guess is the error; and gradient descent nudges every weight downhill, so the error gets smaller.